CNC Procedure 2016_0021

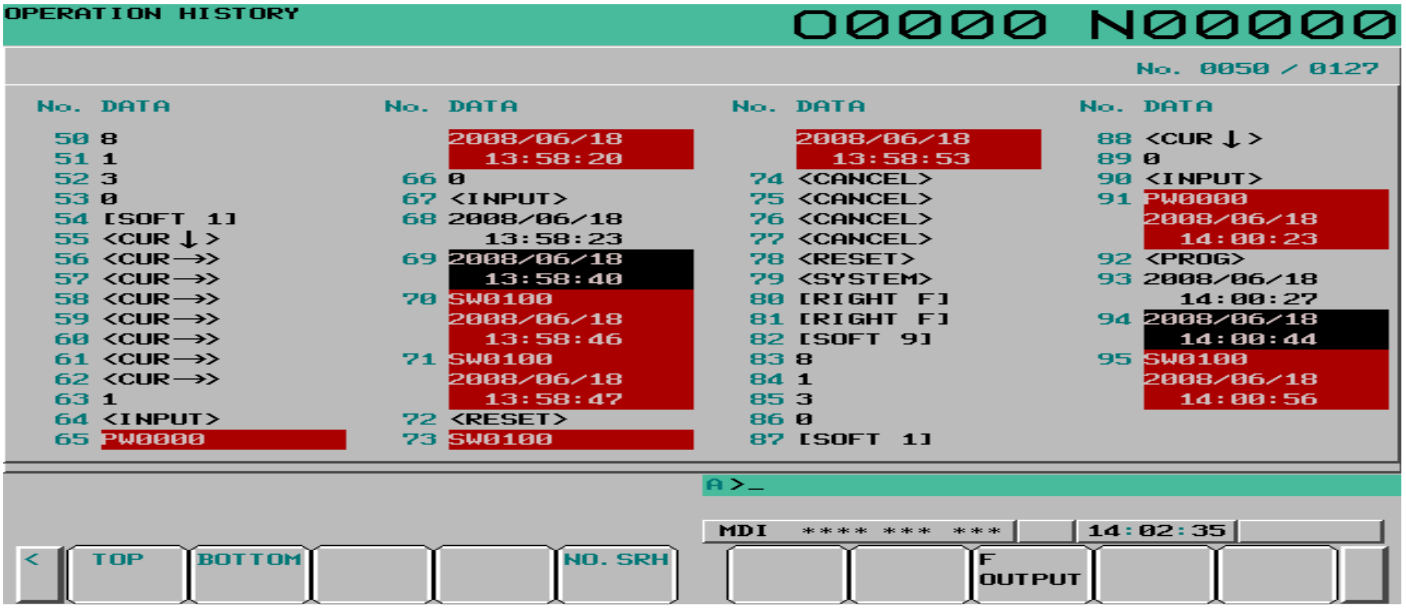
May 5th

2016

Procedure for OPERATION HISTORY CONTENT LIMITING PARAMETERS

Applicable for FS0iF and above CNC's

OPERATION HISTORY:



This function displays a history of the operator's key operations and signal operations made when a failure occurred or an alarm was issued, and also information about alarms.
The following data is recorded:

a Operation history

- i MDI key operations made by the operator
- ii I/O signal (X,Y,G,F,R) on/off switching

b Alarm history

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- i Alarms issued
- ii Modal information in a block executed and coordinates observed when an alarm was issued (Not displayed on the screen)

c Data modification history

- i Modification of tool offset data
  (When bit 0 (HTO) of parameter No. 3196 is set to 1)
- ii Modification of workpiece offset data/extended workpiece offset data/workpiece shift (T series)
  (When bit 1 (HWO) of parameter No. 3196 is set to 1)
- iii Modification of parameters
  (When bit 2 (HPM) of parameter No. 3196 is set to 1)
- iv Modification of custom macro common variable data
  (When bit 3 (HMV) of parameter No. 3196 is set to 1)

d External operator message history and macro message history
(When bit 6 (HOM) of parameter No. 3196 is set to 0)

e Time stamp (time and date)
With some exceptions, history data of the operation history and alarm history can be viewed on the
operation history screen. (The data modification history, external operator message history, and alarm messages are not displayed.)

All recorded history data can be output to external input/output devices.

NOTE

Up to about 8000 items of history data can be recorded if the data includes history data of MDI key operations only. However, because history data varies in size, the maximum number of history data items that can be recorded is not fixed.

Operation History Screen display related PRM are as below:

PRM3106#4 OPH The operation history screen is:
0: Not displayed.
1: Displayed.
PRM3112#3 EAH Messages of the external alarm/macro alarm in alarm or operation history:
0: Not recorded
1: Recorded
Note: This parameter is valid when bit 7 (HAL) of parameter No. 3196 is set to 0.

PRM311~~2~~ 3#6 MS0
PRM311~~2~~ 3#7 MS1 Set the combination of the number of characters and the number of messages to be preserved in the external operator message history.
Parameter Maximum number of characters Number of messages

Parameter		Maximum number of characters	Number of messages
MS0=0	MS1=0	255	8
MS0=1	MS1=0	200	10
MS0=0	MS1=1	100	18
MS0=1	MS1=1	50	32

NOTE
1 This parameter is valid when bit 3 (SOH) of parameter No. 11354 is set to 1.
2 Although up to 255 characters can be specified for each external operator message, you can use the combination of bits 6 (MS0) and 7 (MS1) of parameter No. 3113 to limit the number of characters and select the number of messages to be preserved in the external operator message history.
3 The settings of bits 6 (MS0) and 7 (MS1) of parameter No. 3113 take effect the next time the power is turned on. The external operator message history is erased at that time.
4 Even though you change the settings of bits 6 (MS0) and 7 (MS1) of parameter No. 3113, the alarm PW0000, "POWER MUST BE OFF" is not issued. You must however turn on the power again before the new settings can take effect.
5 If text (such as single-byte katakana or kanji characters) is entered in character code, the number of characters recorded in the external operator message history may be smaller than the maximum number of characters set by bits 6 (MS0) and 7 (MS1) of parameter No. 3113.

PRM3122 Time interval used to record time data in operation history
[Input type] Parameter input [Data type] Word path [Unit of data] min
[Valid data range] 0 to 1440
When history data is recorded within a set time period, the time for each set time period is recorded in the history data. When 0 is set, the specification of a time period of 10 minutes is assumed.

PRM3195#5 HKE A key operation history is:
0: Recorded.
1: Not recorded.

PRM3195#6 HDE A DI/DO history is:
0: Recorded.
1: Not recorded.

PRM3195#7 EKE The [ALL CLEAR] soft key for clearing all history data is:
0: Not displayed.
1: Displayed.

PRM3196#0 HTO A modification history of tool offset data is:
0: Not recorded.
1: Recorded.

PRM3196#1 HWO A modification history of workpiece offset data/extended workpiece offset data/workpiece shift (T series) is:
0: Not recorded.
1: Recorded.

PRM3196#2 HPM A modification history of parameters is:
0: Not recorded.
1: Recorded.

PRM3196#3 HMV A modification history of custom macro common variables is:
0: Not recorded.
1: Recorded.

PRM3196#5 HOA When an external operator message is issued, additional information (modal data, absolute coordinates, and machine coordinates present at the issuance of the external operator message) is:
0: Not recorded in the operation history and message history.
1: Recorded in the operation history and message history.

PRM3196#6 HOM A history of external operator messages and macro messages (#3006) is:
0: Recorded.
1: Not recorded.

PRM3196#7 HAL When an alarm is issued, additional information (modal data, absolute coordinates, and machine coordinates present at the issuance of the alarm) is:
0: Recorded in the operation history and alarm history.
1: Not recorded in the operation history and alarm history.
To record as many alarm history items as possible, rather than detailed alarm information, set 1.

PRM3206#4 PHS Operation history signal selection:
0: Does not interact with parameters.
Operation history signal selection is added or deleted on the operation history signal selection screen. Changing the settings of parameters Nos. 24901 to 24920, Nos. 12801 to 12820, Nos. 12841 to 12860, or Nos. 12881 to 12900 has no effect on operation history signal selection. Changes to the signals of the addresses specified by parameters Nos. 24901 to 24920, Nos. 12801 to 12820, Nos. 12841 to 12860, or Nos. 12881 to 12900 are not recorded in the history.
1: Interacts with parameters.
Operation history signal selection can be performed either on the operation history signal selection screen or by setting parameters.

NOTE
Setting this parameter to 1 reflects the current operation history signal selection data on parameters Nos. 24901 to 24920 and Nos. 12801 to 12900.

PRM11354#2 SAH When the storage capacity for history data is exceeded due to non-alarm history, alarm history will be:
0: Erased.
1: Erased, except the most recent 50 items of history data.

NOTE
The settings of bit 2 (SAH) of parameter No. 11354 and bit 3 (SOH) of parameter No. 11354 will be effective the next time the power is turned on. At this time, all history data (operation history, alarm history, and external operator message history) will be erased.

PRM11354#3 SOH When the storage capacity for history data is exceeded due to data other than external operator message history, external operator message history will be:
0: Erased.
1: Retained.
When this parameter is set to 1, the number of retained external operator message history is determined by the settings of bits 6 (MS0) and 7 (MS1) of parameter No. 3113.
NOTE
With the settings of bit 2 (SAH) of parameter No. 11354 and bit 3 (SOH) of parameter No. 11354, the number of history data items that can be retained varies. The number of history data items that can be recorded as follows:
SAH=0, SOH=0 . . . Approx. 8000 items
SAH=1, SOH=0 . . . Approx. 7400 items
SAH=0, SOH=1 . . . Approx. 7500 items
SAH=1, SOH=1 . . . Approx. 6900 items
(*) The numbers of items above are those if only key operation history is recorded.

PRM24303#4 ODE In the operation history screen, detailed information is
0: Not displayed.
1: Displayed.